Actual torsion boundaries in source claims

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Lean now establishes a direct r=2 discrepancy in FST2010's printed text and an actual excluded-case witness for BK1998. The expanded project gate audits74declarations,0failing,0trusted-native exceptions. Counts include generated equation lemmas; all scientific claims remain unsigned producer assertions. [pcc:unverified]

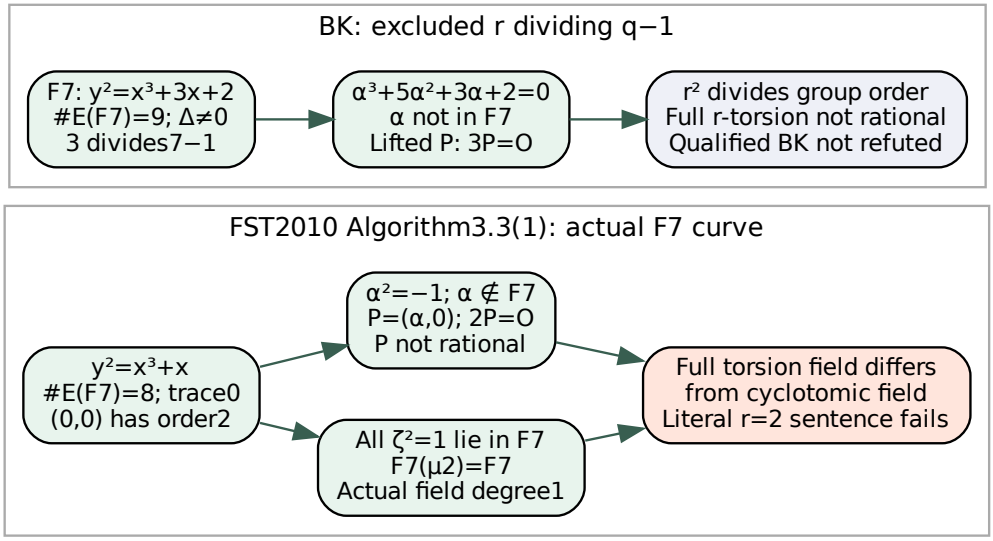


Figure 1: Kernel-checked field and point statements. Every point belongs to Mathlib's actual elliptic group; no hand-written torsion table or assumed Frobenius representation is used.

For E/F7:$y^2=x^3+x$, Lean computes eight rational points and trace0. Thus E meets the source's finite-field supersingularity definition. Also −1 is a nonsquare, so this is an Algorithm3.3(1) output; (0,0) has order2. [pcc:unverified]

FST Definition2.1 defines embedding degree using the field generated by roots of unity. Lean proves in any field extension that all square roots of unity lie in the base field, then computes the actual intermediate-field degree [F7(μ2):F7]=1. Separately it constructs $\alpha^2=-1$ and the actual point (α,0), killed by2 but outside the rational-point image. [pcc:unverified]

The printed paragraph after Algorithm3.3 on p239 assigns embedding degree2 to the subgroup of order2. That literal assertion fails under Definition2.1. The primary PDF page was visually inspected to exclude an extraction error. The distinction is between the full torsion field and the cyclotomic field; the large-prime construction analysis is not refuted. [pcc:unverified]

Formal declaration: FSTBoundaryAudit.fst_algorithm3_3_r_two_counterexample. Unsigned claim: PCC-20261109-2b5092. Source: Freeman–Scott–Teske, DOI10.1007/s00145-009-9048-z, §§2–3, p239. Exact PDF hash/read scope is in ../../sources.json. [pcc:unverified]

For E/F7:$y^2=x^3+3x+2$, Lean computes #E(F7)=9 and $\Delta\neq 0$. A root α of X^3+5X^2+3X+2 has no F7 coordinate. A square-root lift gives an actual point P; the repeated-S3 point theorem proves3P=O. Its x-coordinate proves it is outside the rational-point image. Therefore3|7−1 and9|#E do not force full rational3-torsion. [pcc:unverified]

BK1998's original Theorem1 explicitly excludes r dividing q−1, and its remarks already discuss failure even when r^2 divides the order. Our witness checks this boundary; it does not refute the qualified theorem. The original proof's linear core matches the prior Lean triangular-map proof. The geometric E[r]/Frobenius and Weil-pairing bridges remain open. [pcc:unverified]

Formal declaration: TorsionBoundaryAudit.bk_excluded_boundary_counterexample. Unsigned claim: PCC-20261109-4ebb59. Source: Balasubramanian–Koblitz, DOI10.1007/s001459900040, complete pp141–145 read. [pcc:unverified]

The same module gives an F5 r=2 counterexample to the literal corpus node, using actual point count6 and source trace0. It also derives the corrected k=2 conclusion for odd prime r>2 from actual point-count divisibility and the source supersingular trace definition, **conditional on the explicit Hasse bound**. Hasse itself is not proved here. Claims: PCC-20261109-c29ff9 and PCC-20261109-232086, unsigned. [pcc:unverified]

Primary review also recovered Gaudry2000 and Hildebrand1994. Gaudry's s1/g! statement is a fixed-genus limit for the whole Jacobian; subgroup sampling remains separate. Hildebrand averages over distinct random supports with fixed positive weights and sufficiently large prime modulus; a fixed-hash rho collision constant does not follow. Their higher-genus counting, mixing formalization and requested censuses remain open. Sources: DOI10.1007/3-540-45539-6_2 and DOI10.1007/BF01199265. [pcc:unverified]

The complete metadata inventory preserves all44unread-label nodes from the pinned613-node corpus and all239source records, including170primary-publication records. These are catalogue counts, not completed audits. See ../../UNREAD_BACKLOG.md and ../../paper-audit-backlog.json. [pcc:unverified]

Reproduction: `lake build` and `lake env lean Refutations.lean` under ../../lean/, then the repository proof gate for that project. Receipts: ../../evidence/proof-gate-torsion-fst/. The earlier native census/report is unchanged. This follow-up runs no new empirical experiment or paid cloud job and supplies no second-host census. The original fourteen-item request remains partial; all obligations stay in ../../README.md. [pcc:unverified]

CI metadata corrections preserve original packet bytes and every prior ledger row; no gate is disabled. Main integration remains blocked by the documented divergent ledger histories. Canonical corpus statuses, attack graphs and IC/rho performance graphs remain unchanged pending the applicable review/promotion process. [pcc:unverified]